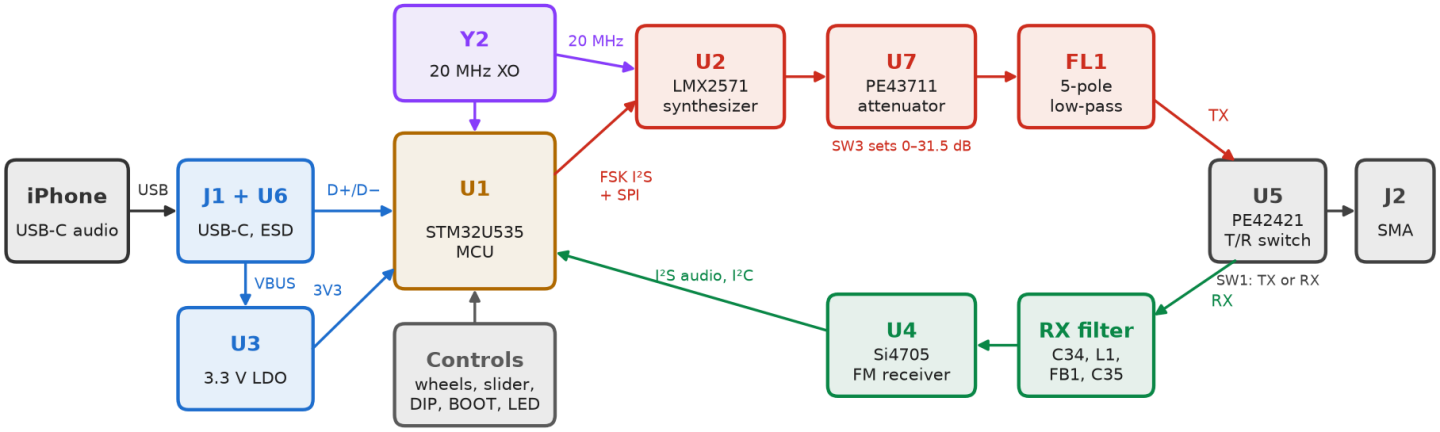
# IphoneFM Rev A — design review

MagSafe FM transmitter / receiver for iPhone (USB-C audio). Prepared for an independent sanity check before the JLCPCB order. 26 Oct 2026. Every claim cites its datasheet page; the datasheets are in the package.

## 1. Overview

**What it is.** A small board that clips to the back of an iPhone and plugs into its USB-C port. The phone sees a USB audio device. In **transmit** it broadcasts the phone's audio on a chosen FM channel (88.1–107.9 MHz, low-power, FCC Part 15 target). In **receive** it tunes an FM station and streams it back to the phone.

**How.** The MCU does all the audio work. The radio is two single-chip parts: a TI LMX2571 synthesizer generates the FM carrier directly (the MCU streams the modulation to it as frequency steps), and a Skyworks Si4705 receives. They share one SMA antenna through an RF switch set by a slider.




| | |
|---|---|
| **Board** | 46.45 × 71.40 mm, 2 layers, 1.6 mm FR-4, 1 oz (JLC0216A stack-up). Ground poured on both layers, 303 vias (123 stitching). |
| **Parts** | 94 placed (45 types), all on the top side; 5 bare test pads underneath. Fully assembled by JLCPCB. |
| **Power** | USB-C 5 V from the phone → AP2112K 3.3 V LDO (600 mA). ~96 mA transmitting, ~62 mA receiving, plus ≤ 15 mA LED. Inside Apple's 100 mA (before USB set-up) / 500 mA limits. |
| **Transmit** | 88.1–107.9 MHz, direct digital FM ±75 kHz, stereo multiplex built in the MCU. Level: LMX2571 −18…+3 dBm, then a 0–31.5 dB DIP-switch attenuator, then a 5-pole low-pass (harmonics est. ≤ −53 dBc). |
| **Receive** | Si4705, I²S digital audio to the MCU, then to the phone as USB audio. |
| **Reference** | One 20 MHz XO (±20 ppm) clocks the MCU, the synthesizer and, through an MCU timer, the receiver. |
| **RF line** | 50 Ω grounded coplanar, 0.92 mm track, 0.20 mm gap. Bottom layer under the RF section is solid ground except two short 3V3 links. |
| **Controls** | 3 thumbwheels (channel digits), TX/RX slider, 6-bit power DIP switch, BOOT button, RGB LED. |
| **Cost** | 2 assembled boards (of 5 made) ≈ $190 before shipping (JLC quote, 26 Oct 2026). |

### Main parts

| Ref | Function | Part (maker) | LCSC | $ / board | JLC stock |
|---|---|---|---|---|---|
| U1 | MCU: USB audio, DSP | STM32U535CEU6 (ST) | C22454224 | 10.08 | 0 → global sourcing |
| U2 | TX synthesizer, direct FM | LMX2571NJKR (TI) | C440388 | 11.32 | 142 |
| U4 | FM receiver, digital audio | Si4705-D60-GM (Skyworks) | C1526114 | 3.11 | 48 |
| U7 | TX attenuator | PE43711B-Z (pSemi) | C470929 | 3.91 | 1,673 |
| U5 | T/R antenna switch | PE42421SCAA-Z (pSemi) | C470893 | 2.92 | 669 |
| Y2 | 20 MHz CMOS reference | OT322520MJBA4SL (YXC) | C725987 | 0.54 | 4,030 |
| U3 | 3.3 V LDO | AP2112K-3.3TRG1 (Diodes) | C51118 | 0.18 | 10,885 |
| U6, D3–D5 | ESD: USB, CC, antenna | USBLC6-2SC6 (ST), TPD1E05U06 (TI), RClamp0521P (Semtech) | C7519, … | 0.36 | deep |
| J1 / J2 | USB-C socket / SMA jack | TYPE-C-31-M-12 (HRO) / BWSMA-KWE-Z001 | C165948 / C496551 | 0.72 | deep |
| RV1–RV3 | Tuning wheels | RK10J11R0A0H (Alps) | C470766 | 2.17 | 908 |

| | | | | | |
|---|---|---|---|---|---|
| SW1 / SW2 / SW3 | Slider / BOOT / DIP | MSK12C02 / EVQ-P7J01P / TDP06H1SBD1 (C&K) | C431540, … | 3.63 | SW3: global sourcing |

Passives: 0402–0805; 16 of the 45 part types are JLC Basic parts. Full BOM, placement file and assembly drawing are in the package.

### Fabrication and assembly

- Our own 1-up panel (KiKit): 6.5 mm rails all round, because parts overhang every edge (wheels, SMA barrel, USB-C, switches); 5 mouse-bite tabs; 3 fiducials.
- DRC: 0 errors. One rule exception: J1's two unplated locating holes sit 0.22 mm from its pads (maker's land pattern; JLC minimum 0.20 mm).
- Placement file corrected per part against JLC's own footprints (rotation and origin), and checked in JLC's preview. JLC engineer placement confirmation is on.
- Gerbers read back with an independent viewer and match the board.

### Firmware plan (not written yet)

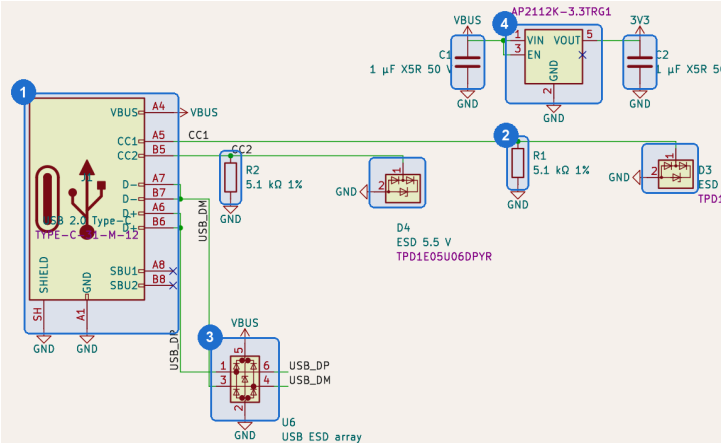
1. **Load:** STM32 ROM USB bootloader (hold BOOT while plugging in); SWD pads as backup.
2. **USB:** Audio Class 2, 48 kHz stereo, asynchronous with feedback (Apple's recommendation).
3. **TX:** MCU builds the FM stereo multiplex (75 µs pre-emphasis, 19 kHz pilot, 38 kHz L−R), corrects the PLL response, interpolates to ~912 kSa/s and streams it to the LMX2571 over SAI in FSK $I^2S$ mode (TI SNAA309). LMX2571 set up over bit-banged MICROWIRE; lock watched on MUXout.
4. **RX:** Si4705 over $I^2C$; its $I^2S$ audio into the second SAI block, back to the phone.
5. **UI:** wheels on the ADC → channel; slider read for mode (it also switches the RF path directly); LED shows state.

## 2. Signal chains

Each chain is drawn on the actual schematic; the numbered badges follow the signal. Page references are to the datasheet in the package.

### Power and USB

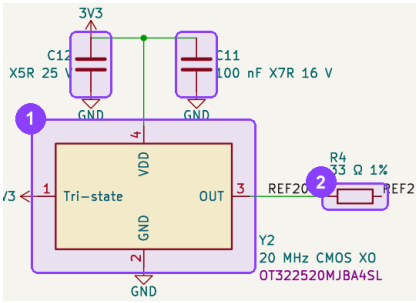
1 J1 USB-C → 2 R1/R2 Rd + D3/D4 ESD → 3 U6 ESD (D+, D−, VBUS) → 4 U3 LDO → 3V3




- 5.1 kΩ Rd on each CC line makes the phone switch on 5 V; no USB-PD needed. USB Type-C R2.5 Table 4-28 p246
- USBLC6 at the connector on D+, D− and VBUS; TPD1E05U06 on each CC. USBLC6-2 p1–2
- AP2112K: 600 mA, stable with ≥ 1 µF; load ≤ ~110 mA. AP2112 p1–2
- Ferrites split the ADC supply (VDDA, FB2) and the receiver supply (FB3) from digital 3V3. DS14217 Fig. 23 p127; Si4705 p16

### Reference clock

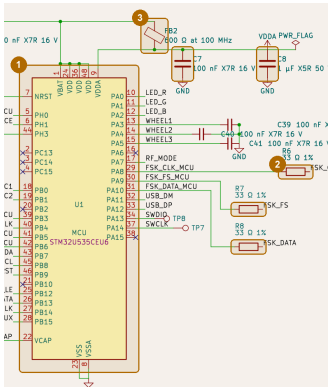
1 Y2 20 MHz XO → U2 OSCin and 2 R4 33 Ω → U1 PH0




- CMOS straight into OSCin, the input TI recommends for best spurs. LMX2571 Table 37 p38, p11
- U1 runs in HSE bypass from the same clock. DS14217 p184–185
- U1 timer makes 32,787 Hz for the Si4705's RCLK (31.1–34.4 kHz allowed). AN332 p37

### MCU

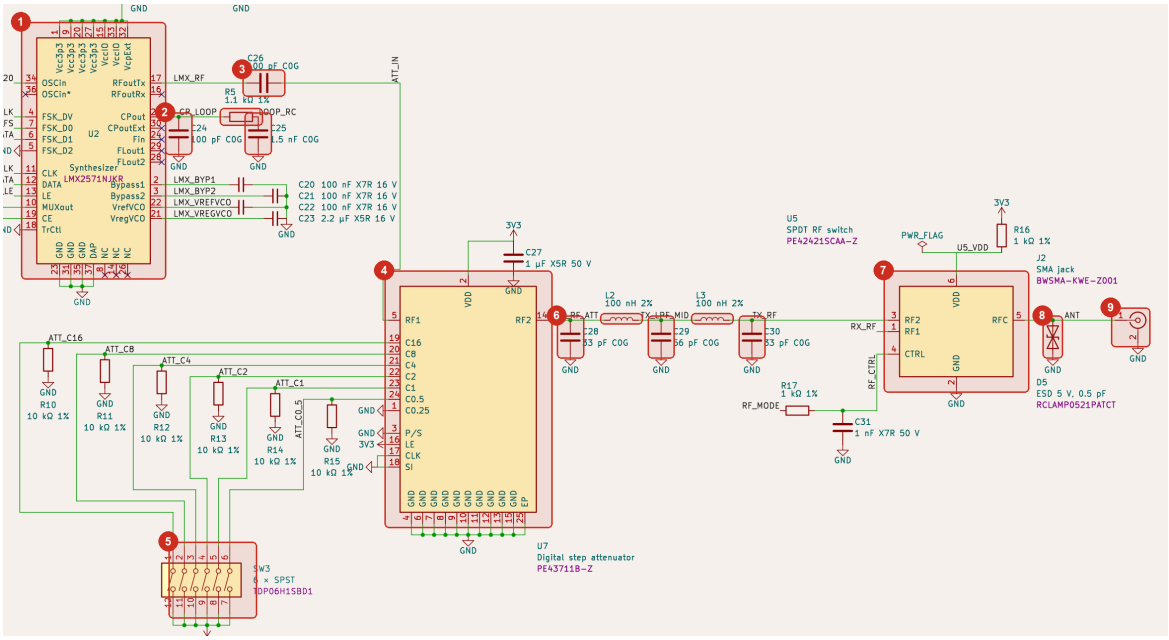
1 U1 STM32U535 → 2 R6–R8 33 Ω FSK I²S · 3 FB2 VDDA



- Bypassing per ST's power scheme; 4.7 µF on VCAP. DS14217 p127
- Pins checked against the alternate-function and bootloader tables. DS14217 Tables 26–28; AN2606 Table 165

### Transmit

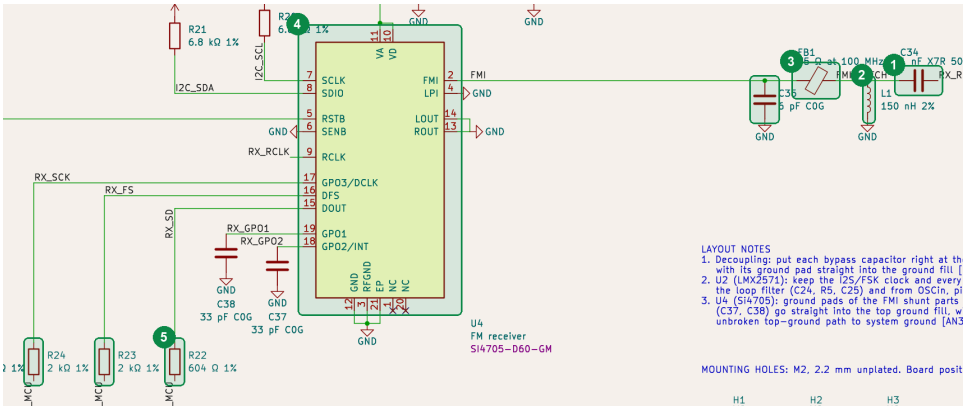
1 U2 LMX2571 → 2 loop filter · 3 C26 DC block → 4 U7 attenuator ← 5 SW3 → 6 FL1 low-pass → 7 U5 switch → 8 D5 ESD → 9 J2



- LMX2571 per TI's typical schematic: 100 nF on every supply pin, 2.2 µF on VregVCO, VcpExt powered. LMX2571 Fig. 62 p44; p53–54
- Loop filter 100 pF / 1.1 kΩ / 1.5 nF → 300–405 kHz loop, ~61° margin; flat to 15 kHz, +0.6 dB at 53 kHz (MCU pre-corrects). Charge-pump current is a firmware setting. LMX2571 p5–6, p35
- Divider ÷56 / ÷48 and two reference settings give ±100 kHz deviation headroom on all 100 US channels. LMX2571 Eq. 4 p35; SNAA309 §6.2
- PE43711 in direct parallel mode from the DIP switch; pull-downs make an open switch read 0. PE43711 p8, Table 4 p6
- FL1: 5-pole 50 Ω low-pass, 23–34 dB at the 2nd harmonic, ≥ 43 dB from 264 MHz. 47 CFR §15.239 / §15.209
- PE42421: ~85 dB isolation keeps the TX carrier off the receiver; the slider drives its CTRL pin directly. PE42421 p3–4

### Receive

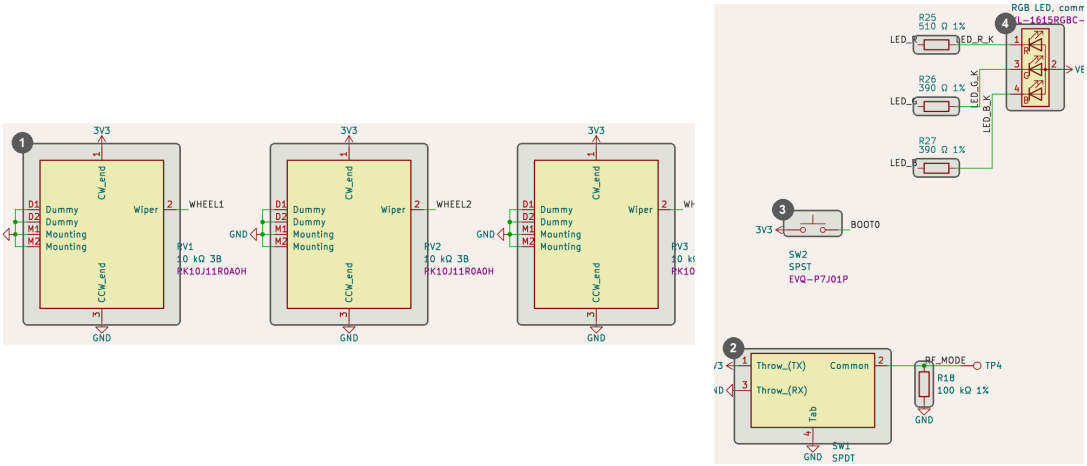
J2 → U5 RF1 → 1 C34 1 nF → 2 L1 150 nH match → 3 FB1 + C35 filter → 4 U4 Si4705 → 5 R22–R24 → U1 SAI




- Front end, emissions filter and I²S series parts as Skyworks specifies. Si4705 p16, Table 12 p18; AN383 p7–9, p37
- U2 is powered down (CE) while receiving, so its carrier can't desensitise U4. LMX2571 p8

### Controls

1 RV1–RV3 → ADC · 2 SW1 → RF_MODE · 3 SW2 BOOT0 · 4 D1 LED




- Wheels are 10 kΩ dividers 3V3–GND, read by the ADC; firmware splits each into 10 digits.
- Slider: 3V3 = TX, GND = RX, 100 kΩ holds RX between contacts; it drives U5 and is read by U1. PE42421 p4
- BOOT0 pulled low; pressed at plug-in → ROM USB bootloader. AN2606 p30
- LED anode on VBUS (blue/green need up to 3.4 V), sunk by U1's 5 V-tolerant open-drain pins. XL-1615 p3–4; DS14217 p132

## 3. What I'd most like checked

1. **Direct FM with the LMX2571.** No published schematic uses it this way; is the loop filter (C24, R5, C25) and the FSK I²S approach sound?
2. **RF layout on 2 layers:** the 50 Ω line, ground stitching and return paths under U2, U5 and U4, and TX-to-RX isolation.
3. **FL1 values** (33 pF, 100 nH, 56 pF, 100 nH, 33 pF) against a square-wave source, and whether its harmonic margin is enough for Part 15.
4. **USB-C sink:** Rd only, no PD controller; current budget against Apple's limits.
5. **Clocking:** one 20 MHz XO for MCU and synthesizer; 32,787 Hz from a timer for the Si4705.
6. **Manufacturing:** the panel and rails, the J1 hole-clearance exception, the JLC placement corrections.
7. Anything that would stop it working on first power-up.

### In the package

**Schematic** IphoneFM-RevA-schematic.pdf · **For JLC** IphoneFM-RevA-gerbers.zip (panel), -BOM.csv, -CPL.csv · **Bench** IphoneFM-RevA-assembly.pdf · **Board** IphoneFM-RevA-board-top.png, -bottom.png, kicad/ project · **Datasheets** datasheets/ (every part datasheet and app note cited; the USB Type-C spec and Apple's accessory guidelines are public)